

PROJECT TITLE: OptiCrop: Smart Agricultural Production Optimization Engine

1. PROBLEM-SOLUTION MAPPING MATRIX:

OptiCrop directly maps standard identified agricultural field problems with data-driven predictive computing solutions:

- * Identified Problem A: Over-reliance on random localized guesswork leading to heavy crop failures.

- * Computational Fit: An intuitive analytical Flask dashboard requiring simple direct numbers to process instant optimized answers.

- * Identified Problem B: Unpredictable weather shifts (uncontrolled temperature, sudden humidity spikes, variable rainfall metrics).

- * Computational Fit: Training a multi-variable Machine Learning classifier engine that understands non-linear historical environmental correlations.

- * Identified Problem B: Highly expensive, long-delayed, and inaccessible laboratory soil testing frameworks for basic field updates.

- * Computational Fit: A lightweight software optimization engine requiring zero installation that renders optimal production recommendations instantly under 2.0 seconds.